FIITJEE

ALL INDIA TEST SERIES

OPEN TEST

JEE (Advanced)-2021

PAPER –1

TEST DATE: 24-01-2021

Time Allotted: 3 Hours **Maximum Marks: 198**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

  Section-A (01 – 06, 19 – 24, 37– 42) contains **18 multiple** choice questions which have **ONLY ONE CORRECT ANSWER**. Each question carries **+3 marks** for correct answer and **–1 mark** for wrong answer.

  Section-A (07 – 12, 25 – 30, 43 – 48) *this section contains* ***18 multiple*** *choice questions.*
  Each question has FOUR options. ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct answer(s).*
  For each question, choose the option(s) corresponding to (all) the correct answer(s)
  Answer to each question will be evaluated according to the following marking scheme:

  Full Marks **: +4** If only (all) the correct option(s) is (are) chosen:
  Partial Marks **: +3** If all the four options are correct but ONLY three options are chosen;
  Partial Marks **: +2** If three or more options are correct but ONLY two options are chosen and both of which are correct;
  Partial Marks **: +1** If two or more options are correct but ONLY one option is chosen and it is a correct option;
  Zero Marks **: 0** If none of the options is chosen (i. e. the question is unanswered);
  Negative Marks **: –2** In all other cases.

  Section-C (13 – 18, 31– 36, 49 – 54) *contains* ***18 Numerical*** *answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***0 marks*** *for wrong answer.*

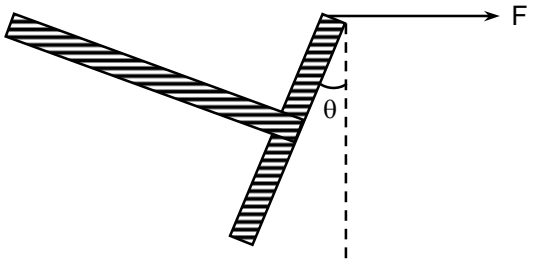
Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A T-shaped rod with its plane lying on a smooth horizontal surface is being pulled without rotation by a constant horizontal force F as shown in the figure. The two straight rods have equal lengths and equal masses. Then the value of angle θ is:



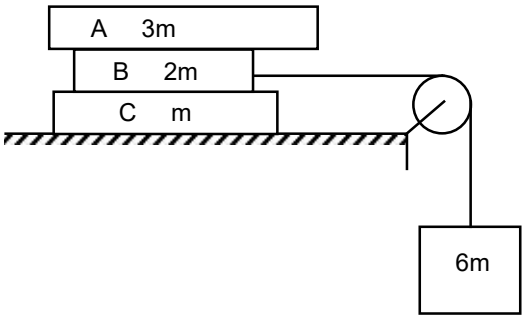
(A) $\tan^{-1}\left(\frac{1}{2}\right)$

(B) $\tan^{-1}(2)$

(C) 0°

(D) 60°

2. In the shown figure all surfaces are rough. Block B is attached with a load of mass 6m through an ideal string passing over an ideal pulley. Coefficient of friction is μ = 0.2 between all the surfaces. The correct option is:



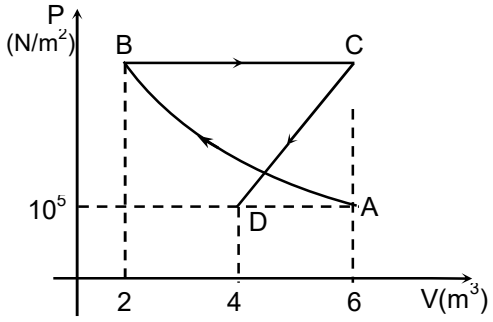
(A) friction force on C due to B is mg

(B) friction force on A is zero

(C) acceleration of block B is $\frac{55}{11}$

(D) Load remains at rest.

3. Two moles of an ideal monoatomic gas is taken through the process as shown in figure. The process is
$A \rightarrow B$ Adiabatic
$B \rightarrow C$ isobaric
$C \rightarrow D$ Linear relation between P and V.
The net change in internal energy of the gas is:



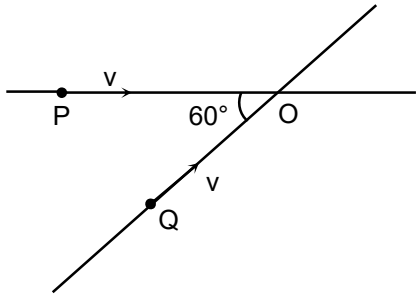
(A) 100 kJ

(B) –200 kJ

(C) –300 kJ

(D) 500 kJ

4. Two cars P and Q are moving with equal speed v = 40 km/h on the two roads which are intersecting at 60°at point O as shown in the figure. Initially, PO is 80 km and QO is 50 km. The minimum separation between them will be:



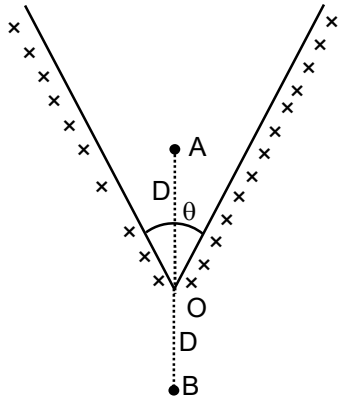
(A) 15 km

(B) $15\sqrt{3}$ km

(C) 5 km

(D) 30 km

5. In the shown figure an infinitely long V shaped, uniformly charged wire is placed. Linear charge density on it is λ. The ratio of the electric field intensity (magnitude) at A and B is: (AB bisects the angle and points A and B are at equal distances from the point O)



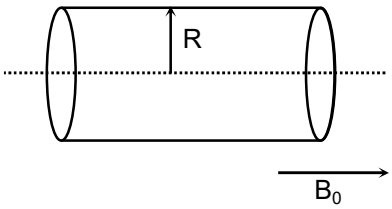
(A) 2

(B) 1

(C) $\cos\left(\frac{\theta}{2}\right)$

(D) $\tan\left(\frac{\theta}{2}\right)$

6. A uniformly charged nonconducting cylinder of mass M, radius R and charge Q is shown in the figure. A uniform magnetic field B_0 parallel to the axis of the cylinder is present in the region. Find the angular velocity of the cylinder when the magnetic field is suddenly switched off.



(A) $\frac{Q}{2M}B_0$

(B) $\frac{2Q}{M}B_0$

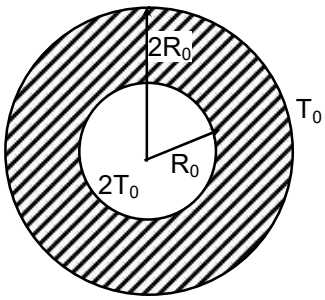
(C) $\frac{Q}{M}B_0$

(D) $\frac{3Q}{2M}B_0$

(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

7. A spherical shell has variable thermal conductivity $K = K_0x$, where x is the distance from the centre of the shell and K_0 is a positive constant. The inner and outer radii of the spherical shell are R_0 and $2R_0$ respectively as shown in the figure. Temperature of inner and outer surfaces are $2T_0$ and T_0 respectively. Then choose the correct option(s).



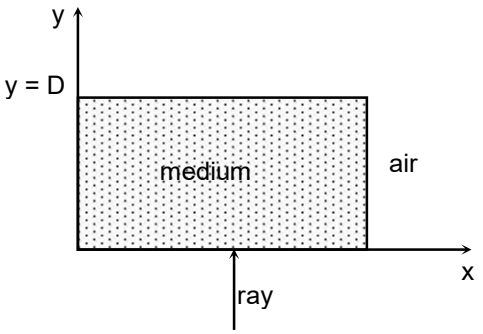
(A) Rate of heat flow is $\frac{8\pi K_0 R_0^2}{3}T_0$

(B) Temperature at radial distance $x = \frac{3R_0}{2}$ is $\frac{3}{2}T_0$

(C) Isothermal surfaces are spherical

(D) The temperature at $x = R_0\sqrt{2}$ is $\frac{4T_0}{3}$

8. A transparent slab is placed in x-y plane as shown in the figure. The refractive index of the medium is given as $\mu = \mu_0\left[1+\alpha\frac{y}{D}\right]$, where $\alpha = 2$. A ray is incident perpendicularly on the medium as shown in the diagram. Then



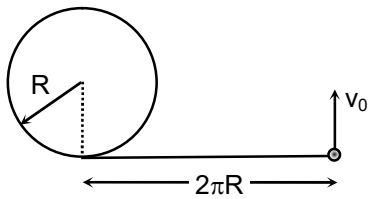
(A) Trajectory of the ray inside the medium is parabola
(B) Trajectory of the ray inside the medium is straight line
(C) The time taken by the ray to cross the medium is $\frac{2\mu_0 D}{c}$
(D) Equivalent optical path length of medium along trajectory of the ray is $2\mu_0 D$.

9. When a monochromatic light from a point source falls on a photo sensitive surface, the maximum kinetic energy of photoelectrons emitted is K_{max}. If the distance between the source and the surface is doubled:
(A) Number of photoelectrons coming out of the surface becomes double.
(B) Number of photoelectrons coming out of the surface becomes one-fourth of the initial value.
(C) K_{max} gets doubled
(D) K_{max} remains same.

10. A particle of mass m and charge q is projected from origin along the positive x-axis with an initial velocity v_0. In this region a uniform electric field $\vec{E} = E_0\hat{i}$ and a uniform magnetic $\vec{B} = B_0\hat{j}$ are present. Then choose the correct option(s).
(A) motion of the particle is in z-x plane
(B) motion of the particle is in x-y plane
(C) maximum velocity of the particle along x-axis is v_0
(D) maximum velocity of the particle along x-axis is $\sqrt{v_0^2+\frac{E_0^2}{B_0^2}}$

11. When a small object is placed near the principle axis of a lens:
(A) Real object always has virtual image in concave lens
(B) Real object always has real image in convex lens
(C) Virtual object always has real image in convex lens
(D) Virtual object always has virtual image in concave lens

12. A particle of mass m is connected with the fixed vertical cylinder with the help of a string of length $\ell_0 = 2\pi R$ as shown in the figure. The particle is given a horizontal velocity v_0 perpendicular to the string and it moves on a smooth horizontal surface. Then choose the correct option(s).



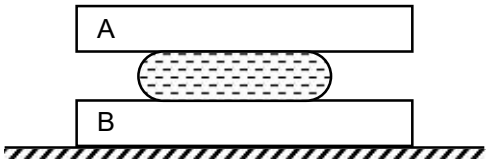
(A) Speed of the particle increases continuously
(B) Speed of the particle remains constant
(C) Tension in the string increases continuously
(D) Time after which the particle will collide with the cylinder is $\frac{2\pi^2 R}{v_0}$

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

13. A hollow cone floats with its axis in vertical direction, having its vertex downward with one third of its height dipped in water. When this cone is filled up to half depth with a liquid, it floats in the water fully submerged. Find the relative density $\left(\frac{\rho}{\rho_w}\right)$ of the liquid filled in it.

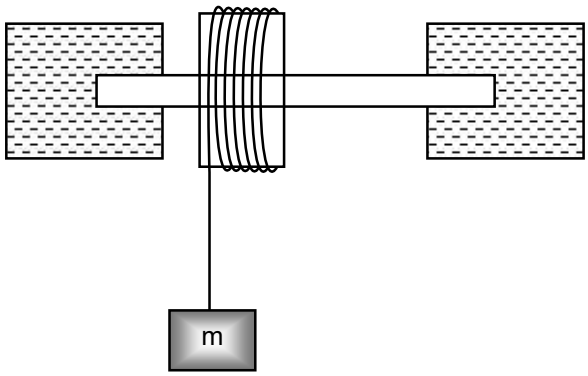
14. A drop of mercury is present between two glass plates each of mass 50 gm. Separation between the plates is negligible as compared to spread of the mercury between the plates. What should be the mass of a block (in kg) which should be placed over the plate A so that the separation between the plates becomes half of the initial separation. Assume contact angle between mercury and glass is 180°.



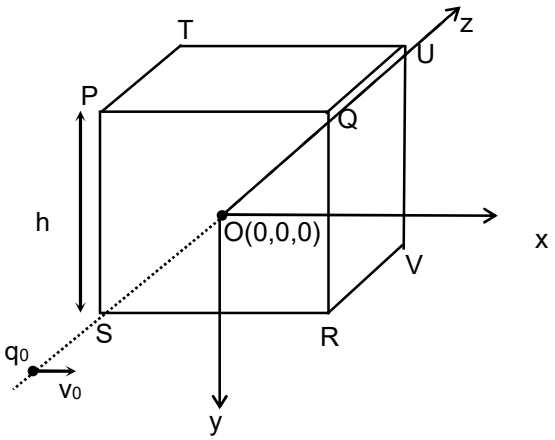
15. In the nuclear reaction ${}_2He^4 + {}_4Be^9 \rightarrow {}_6C^{12} + {}_0n^1$, kinetic energy of the α-particle is 5.32 MeV. The Q-value of this reaction is Q = 5.80 MeV. The neutron goes perpendicular to the direction of the incoming α-particle. Find the kinetic energy (in MeV) of C^{12} produced.

16. In a photoelectric experiment work function of the emitter is 10 eV and a monochromatic light of wavelength 124 Å is falling over it. The accelerating voltage applied across emitter and the metallic collector is 10 kV. Find the minimum wavelength (in Å) of the radiation emitted from the collector. (Take hc = 12400 eV-Å)

17. A block of mass m = 2kg is attached with a string which is wrapped over a pulley of radius R = 10 cm. The ends of the pulley are symmetrically supported by viscous liquids of equal amount. Heat capacity of each of the viscous liquids is H = 100 J/°C. It is found that block is moving down with a uniform velocity, v_0 = 2.5 m/s. Find the rate of increase of temperature of the viscous liquids in °C/s. Assume no loss of heat to the surrounding and the axle of the pulley does not take any heat. (Take g = 10 m/s^2)



18. A cuboid of mass 4 kg is kept in a gravity free space. Its sides are parallel to the coordinate axes as shown in the figure. The edges parallel to the x-axis and z-axis have lengths $a\sqrt{3}$ where as the one parallel to the y-axis is of the length h. There is a uniform electric field $\vec{E} = ka\hat{j}$ N/C (where k=1 N/C-m) and a uniform magnetic field $\vec{B} = \frac{4\pi}{3}\hat{j}$ Tesla. A particle having charge q_0 and mass m is given an initial velocity $\vec{v}_0 = \frac{\pi a}{3}\hat{i}$ m/s from point (0, 0, $-a$). It strikes the cuboid with coefficient of restitution e = 1/2. Find the angular momentum of the whole system after collision (about origin) in kg-m^2/s. (Take m = 1 kg, q_0 =0.25 C, a=4 meters , π =3, $\sqrt{6} = 2.45$ and h is sufficiently large)



O is the centre of the face PQRS

Chemistry

PART – II

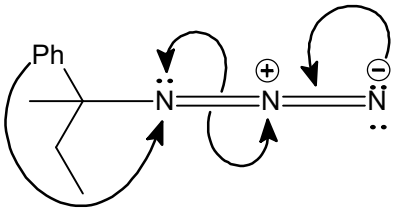
SECTION – A
(One Options Correct Type)

This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

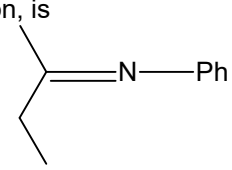
19. Total number of isomers of a compound having molecular formula, $K_3Fe(C_6N_6)$ is:

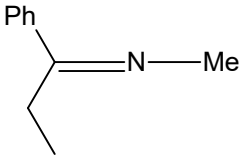
(A) 7
(B) 0
(C) 9
(D) 10

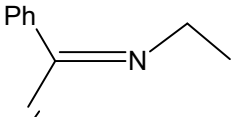
20.

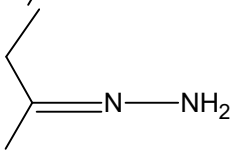


The major electrically neutral product that can be formed in the above mentioned curved arrow notation, is

(A)


(B)


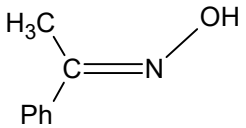
(C)


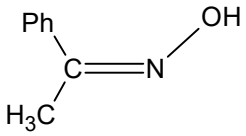
(D)


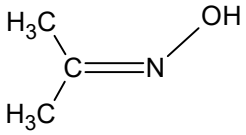
21. Ketone 'P" on treatment with NH_2OH at pH = 5 gives compound (A).

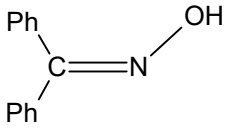
$$\text{Compound}(A) \xrightarrow[\Delta]{H_2SO_4} \underset{(\text{Major})}{N-\text{methyl benzamide}}$$

So, correct structure of 'A' is:

(A)


(B) 

(C) 

(D) 

22. 1 mole of $H_2(g)$ and 1 mole of $I_2(g)$ are taken in a container at 400 K and the following equilibrium is allowed to establish:

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

The equilibrium mixture required 100 mL of 10 M hypo solution up to starch end point (i.e. blue colour of starch indicator is disappeared). So, the dissociation constant per mole of HI at 400 K will be:

(A) 0.25
(B) 0.50
(C) 4
(D) 2

23. 36 mL of pure water takes 100 seconds to evaporate at $100^\circ C$ and 1 atm, from a vessel and heater connected to an electric source which delivers 806 watt. Now, select the only INCORRECT statement about this process:

(A) Enthalpy of vapourisation of $H_2O(\ell)$ is 40.3 kJ mol^{-1}

(B) Entropy change of the system is 216.08 J K^{-1}

(C) Internal energy change of the system is 74.3 kJ

(D) Change in free energy of $H_2O(\ell)$ is -40.3 J mol^{-1}

24. Among the reactions given below for B_2H_6, the one which does not take place, is:

(A) $B_2H_6 + HCl \longrightarrow B_2H_5Cl + H_2$

(B) $B_2H_6 + 2NH_3 \xrightarrow{\Delta} B_3N_3H_6$

(C) $B_2H_6 + 2N(CH_3)_3 \longrightarrow 2(CH_3)_3NBH_3$

(D) $B_2H_6 + 6C_2H_4 \xrightarrow{H_3O^+} 6C_2H_5OH + 2B(OH)_3$

(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

25. Which of the following statement(s) is/are correct?

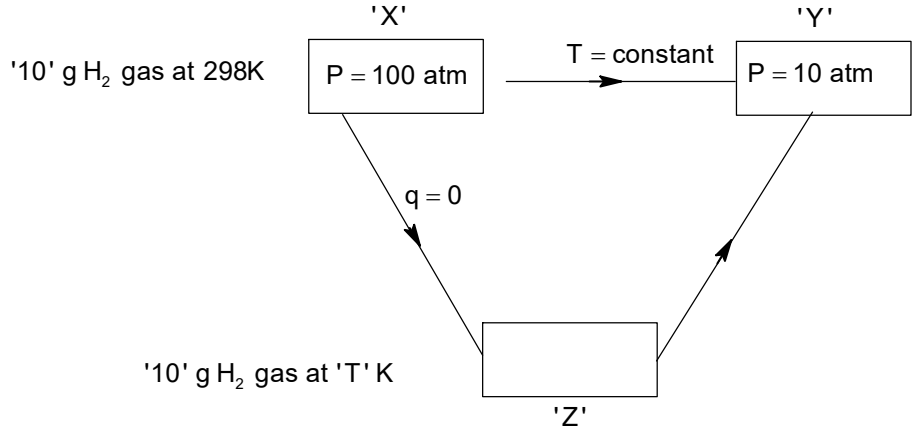
(A) Highly pure dilute solution of sodium in liquid ammonia is red in colour

(B) Sodium nitrate decomposes above $800^\circ C$ and gives $N_2(g)$ as one of the product

(C) When sodium metal is burnt in excess of air, it forms mainly Na_2O_2 and traces of Na_2O

(D) On addition of catechol to aqueous solution of boric acid, its pH increases

26. Consider the following process (each step involves a reversible change)



Select the correct choice(s):

(A) $\Delta S_{X\to Z} = 0$

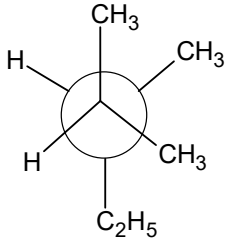
(B) $\Delta S_{X\to Z\to Y} = 0$

(C) $\Delta S_{X\to Y} = 95.7\ JK^{-1}$

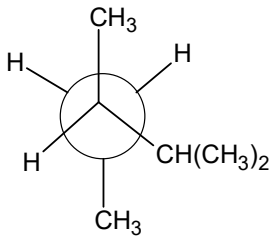
(D) $\Delta S_{X\to Y} = \Delta S_{Z\to Y} = 95.7\ JK^{-1}$

27. Which of the following represents identical configuration of the molecule?

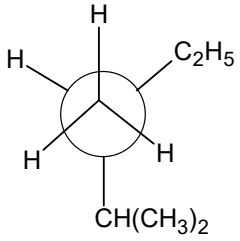
(A)



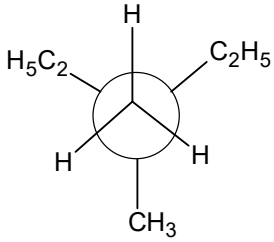
(B)



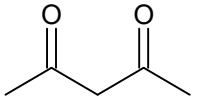
(C)

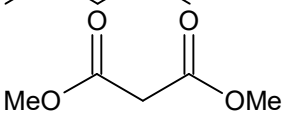


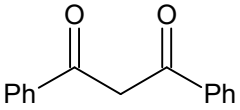
(D)

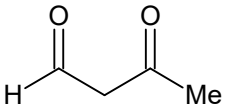


28. Which of the following has/have enol content higher than methyl -3-oxobutanoate?

(A) 

(B) 

(C) 

(D) 

29. Which of the following species has/have only one corner shared per tetrahedron unit?

(A) $Si_2O_7^{6-}$

(B) $Cr_2O_7^{2-}$

(C) $S_2O_7^{2-}$

(D) $Si_3O_9^{6-}$

30. If degeneracy of first excited state of chlorine atom is 'x', degeneracy of second excited state of helium atom is 'y' and degeneracy of second excited state of hydrogen atom is 'z', then correct value of x, y and z is/are (ignore the nuclear spin)

(A) x = 5

(B) y= 3

(C) z = 3

(D) z = 9

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. If total number of exact $109°28'$ bond angles in a molecule of tetramethyl silane be 'x' and total number of lone pair of electrons in a molecule of tear gas (Nitro chloroform) be 'y', then find the value of $\frac{y}{(x-1)}$.

32. Under strongly acidic conditions, one mole nitrobenzene on electrolytic reduction gives a product (x) and consumes 'n' Faraday of charge while under mild acidic conditions, one mole nitrobenzene, on electrolytic reduction gives product (y) and consumed 'm' Faraday of charge. So, the value of $\frac{m}{n}$ is

33. Consider the following compounds:

$O_2(\ell), O_3(g), \left[Cu(NH_3)_4\right]SO_4, K\overset{III}{Fe}\left[\overset{II}{Fe}(CN)_6\right]$

$K\overset{II}{Fe}\left[\overset{III}{Fe}(CN)_6\right], N_2O_3(\ell), HgI_2, CuO, Cu_2O, (C_2H_5)_3BO_3,$

KO_3, Pb_3O_4, Ag_2CrO_4, CaC_2O_4

Let, number of compound(s) having blue or pale blue colour be 'x'
Number of compound(s) having red colour be 'y'
and number of compound(s) having neither red nor blue or pale blue be 'z'.

Then find the value of $\left(\frac{x+z}{y}\right)$.

34. Consider the following reaction:

$$A \underset{k_b=0.02\,sec^{-1}}{\overset{k_f=3\,min^{-1}}{\rightleftharpoons}} B$$

where k_f and k_b are the rate constants of forward and backward reactions respectively. So, half-life time (in seconds) of 'A' will be:

35. Five gram of an organic compound was analysed by Kjeldahl's method. Ammonia evolved was absorbed in 75 mL of $\frac{M}{5}H_2SO_4$ solution. Un-used acid required 50 mL of $\frac{N}{5}NaOH$ for complete neutralization. Hence, the mass percentage of nitrogen in the organic compound is…..

36. If the slope of compressibility factor (Z) versus pressure (P) curve of one mole of a real gas is constant and equal to $\left(\frac{\pi}{492.6}\right)atm^{-1}$ at 300 K and very high pressure; then calculate the radius of the gas molecule in $\left(A^\circ\right)$.

[Given : Avogadro's number $= 6\times10^{23}$, $R = 0.0821\,L\,atm\,mol^{-1}\,K^{-1}$, $3\sqrt{\frac{47}{3}} = 2.50$]

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. If the equation $x^4 - 4\sqrt{3}\tan\theta x^3 + 16\sqrt{3}\tan\theta x - 16 = 0$ has at least two positive roots, then

(A) $\theta \in \left(n\pi - \frac{\pi}{2}, n\pi + \frac{\pi}{3}\right]; n \in z$

(B) $\theta \in \left[n\pi - \frac{\pi}{3}, n\pi + \frac{\pi}{3}\right]; n \in z$

(C) $\theta \in \left[n\pi + \frac{\pi}{6}, n\pi + \frac{\pi}{2}\right); n \in z$

(D) $\theta \in \left(n\pi - \frac{\pi}{2}, n\pi + \frac{\pi}{6}\right]; n \in z$

38. Let R_1 and R_2 be the radii of the circles C_1 and C_2 touching the ellipse $3x^2 + 4y^2 = 7$ at the point P(1, 1) and also touching x-axis; ($R_1 > R_2$), then the point Q on the circle C_2 which is farthest from the circle C_1 is

(A) $\left(\frac{1}{3}, \frac{2}{9}\right)$

(B) $\left(\frac{1}{3}, \frac{1}{9}\right)$

(C) $\left(\frac{2}{3}, \frac{1}{15}\right)$

(D) $\left(\frac{1}{5}, \frac{1}{15}\right)$

39. Let y(x) be a solution of the differential equation $x(1 - x^3)^2\, dy + ((2x^3 y + y)(1 - x^3) - 3x^2)dx = 0$.

If $y\left(2^{1/3}\right) = -\frac{1}{2^{4/3}}$, then $\int y(x)dx$ is

(A) $\ln\left|\frac{x^3}{1-x^3}\right| + c$

(B) $\frac{1}{6}\ln\left|\frac{1-x^3}{x^3}\right| + c$

(C) $\frac{1}{6}\ln\left|\frac{x^3}{1-x^3}\right| + c$

(D) $\frac{1}{2}\ln\left|\frac{2x^3}{1-x^3}\right| + c$

40. Consider a hyperbola having its foci at (–4, –1) and (6, 9) and the foot of the perpendicular from the focus (6, 9) upon a tangent to the hyperbola is (5, 7), then the point of contact of tangent is
(A) (1, 9)
(B) $\left(\frac{13}{2}, \frac{7}{2}\right)$
(C) $\left(6, \frac{13}{2}\right)$
(D) $\left(\frac{13}{2}, 6\right)$

41. Let $f(x) = \min\left\{1, 1+x^5, |x^2-5x+6|, |x-4|\right\}$; $x \in R$, then
(A) f(x) has 8 local extremum and is non differentiable at 7 points
(B) f(x) has 7 local extremum and is non differentiable at 6 points
(C) f(x) has 5 local extremum and is non differentiable at 6 points
(D) f(x) has 6 local extremum and is non differentiable at 7 points

42. Let $\vec{a}$ and $\vec{b}$ be two non-collinear vectors such that $|\vec{a}| = |\vec{b}| = \sqrt{2}$ and $|\vec{a} \times \vec{b}| = |\vec{a} - \vec{b}|$. If $\vec{r}$ is any vector such that $|\vec{r}| = \sqrt{\frac{15}{2}}$ and $\vec{r} \times (\vec{a} \times \vec{b}) = 2\vec{a} - 3\vec{b}$, then the value of $\left[\vec{r}\ \ \vec{a}\ \ \vec{b}\right]$ is equal to
(A) 1
(B) $\sqrt{2}$
(C) 2
(D) $\sqrt{3}$

(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

43. Let A be the set of 2×2 matrices given by $A = [a_{ij}]$; where $a_{ij} \in \{0, 1, 2, 3, 4, 5\}$ such that $a_{11} + a_{12} + a_{21} + a_{22} = 5$, then
(A) the number of matrices in A which are invertible are 32
(B) the number of matrices in A which are symmetric or skew symmetric are 8
(C) the number of matrices in A which are symmetric or skew symmetric such that Det A is odd are 6
(D) The total number of matrices in A are 56

44. Mr. X has 5 gel pen namely P_1, P_2, P_3, P_4, P_5. He randomly selects one of the four pens which was not used in the previous examination (test). If for his first examination (test) the gel pen used was P_1 and P(n) denotes the probability that he chooses P_1 in the nth test as well then
(A) $P(5) = \frac{13}{64}$
(B) $P(6) = \frac{51}{256}$
(C) $\frac{1}{4} < P(n) < \frac{1}{3}\ \ \forall\ n \geq 3$
(D) $P(n) < \frac{1}{4}\ \ \forall\ n \geq 5$

45. Consider the curves $S_1 \equiv y^2 - 4k_1x = 0$, $k_1 > 0$, $S_2 \equiv x^2 - 4k_2y = 0$, $k_2 > 0$ and $S_3 \equiv x^2 + y^2 + 10y - 27 = 0$. A line $L \equiv 2y + 3x + 36 = 0$ touches $S_3 = 0$, $S_1 = 0$ and $S_2 = 0$ at P, Q and R respectively such that $PQ = PR = \sqrt{1053}$. Moreover S_1 and S_2 intersect at point S (other than origin)

(A) tangents drawn at the extremities of the chords $x - k_1 + \lambda(y) = 0$, (λ is parameter) of the parabola $S_1 = 0$ intersect on the line $x = -8$

(B) tangents drawn at the extremities of the chords $x + \mu(y - k_2) = 0$; (μ is parameter) of the parabola $S_2 = 0$ intersect on the line $y + 8 = 0$

(C) the length of focal chord of the curve $S_1 = 0$ passing through the point S is 1875/16

(D) the points on the curve $S_2 = 0$ at which the normals drawn intersect at S are given by (–16, 8) and (–32, 32)

46. Let $S_1 = \{z \in c ; z + \bar{z} + 1 = |z - 1|\}$, $S_2 = \{z \in c ; |z - 1| + |z + 3| = 6\}$ and $S_3 = \{z \in c : \text{Re } z > 0, \text{Im}(z) \le -2\}$, then which of the following option(s) are incorrect?

(A) The curves S_1 and S_2 cannot intersect orthogonally

(B) If $S = S_1 \cap S_2 \cap S_3$, then number of elements in S is zero

(C) All the chords of the curve $S_1 = 0$ which subtend right angle at origin always pass through the point (–2, 0)

(D) The area of triangle formed by pair of straight lines $3(x + 1)^2 - y^2 = 0$ and the tangent to the curve S_1 at (1, 3) is $\sqrt{3}$ units

47. Let $T_n = \sum_{k=0}^{n-1} \frac{\left(k^3n - k^4 - k^2n^2 + 2020n^4\right)}{n^5}$ and $S_n = \sum_{k=1}^{n} \frac{\left(k^3n - k^4 - k^2n^2 + 2020n^4\right)}{n^5}$, then

(A) $T_n > 2020$, $S_n < T_n$

(B) $S_n > 2020$, $S_n > T_n$

(C) $T_n > 2019$, $S_n < T_n$

(D) $\left[\lim_{n\to\infty} S_n\right] = 2019$ (where [.] denotes greatest integer function)

48. Let f(x) be a function defined as $f : \left[0, \frac{\pi}{2}\right) \to R$ satisfying $\int e^{f(x)+\ln\cos x}\left(x - 2\sin 2x - x\tan^2 x\right)dx = e^{f(x)+\ln\cos x}\cos 2x + c$; f(0) = 0 and g(x) be a function such that g(x) (1 + f(x)) = x, then

(A) f(x) = 1 has no solution

(B) g(x) has one local extremum in $\left(0, \frac{\pi}{2}\right)$

(C) g(x) has exactly one local minimum in $\left(0, \frac{\pi}{2}\right)$

(D) if $h(x) = f^{-1}(x)$; (inverse function of f(x)), then the value of $\frac{d}{dx}(h(x))$ at $x = \frac{\pi}{4}$ is $\frac{2}{\pi + 2}$

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. Let $S_n = \sum_{k=0}^{n} \left({}^{n+k}C_k\right)k$. If $S_{100} = \frac{A!}{B!}\left(\frac{1}{101!} - \frac{1}{102!}\right)$, then the value of A + B is

50. Quadrilateral ABCD is inscribed in a circle with AD as diameter. If AD = 10, AB = BC = 5 and the inradius of triangle BAC = $\frac{A\sqrt{3} - B}{2}$; A, B $\in$ N, then the value of A + B is

51. If $I = \int_0^{\pi/3} \left(\log_2\left(1+\sqrt{3}\tan x\right)\right)\left(\pi x - 3x^2\right)^2 dx$, then the value of $\frac{3240\, I}{\pi^5}$ is

52. Let $S_1(n) = 1^2 + 3^2 - 5^2 - 7^2 + 9^2 + 11^2 - 13^2 - 15^2 + ... + 4n$ terms
$S_2(k) = 1 + 5 + 11 + 19 + ...$ k terms, S = {1, 2, 3, 4, ….. 100}.
If $S_2(40) + S_1(n) < 0$, then number of values in S, then n can accept is/are

53. In an acute angled triangle ABC ; D, E, F are the feet of the altitudes drawn from the vertices A, B and C respectively on the opposite sides. If inradius of ΔDEF is one-fourth of circumradius of ΔABC and the circumradius of ΔABC is $\sqrt{3}$, then the perimeter of ΔABC is

54. In a rectangular parallelepiped of edges a, b, c. Let ℓ_1, ℓ_2, ℓ_3 be the shortest distance between an edge and a solid diagonal not intersecting it. If $\frac{a+b+c}{abc} = 96$, then the minimum value of $\frac{1}{\ell_1^2} + \frac{1}{\ell_2^2} + \frac{1}{\ell_3^2}$